Activity - 3

1. Find the value of k for the given system of linear equations satisfying the condition below:
   (i) $2x + ky = 1;\ 3x - 5y = 7$ has a unique solution
   (ii) $kx + 3y = 3;\ 12x + ky = 6$ has no solution
   (iii) $(k-3)x + 3y = k;\ kx + ky = 12$ has infinite number of solution
2. Find the value of a and b for which the given system of linear equation has infinite number of solutions $3x - (a+1)y = 2b - 1, \quad 5x + (1-2a)y = 3b$

Activity - 4

For the given linear equations, find another linear equation satisfying each of the given condition

Given linear equation	Another linear equation		
	Unique Solution	Infinite many solutions	No solution
$2x+3y = 7$	$3x+4y = 8$	$4x+6y = 14$	$6x+9y = 15$
$3x-4y = 5$			
$y-4x = 2$			
$5y-2x = 8$			

Exercise 3.14

Solve by any one of the methods

1. The sum of a two digit number and the number formed by interchanging the digits is 110. If 10 is subtracted from the first number, the new number is 4 more than 5 times the sums of the digits of the first number. Find the first number.
2. The sum of the numerator and denominator of a fraction is 12. If the denominator is increased by 3, the fraction becomes $\frac{1}{2}$. Find the fraction.
3. ABCD is a cyclic quadrilateral such that $\angle A = (4y + 20)^\circ$, $\angle B = (3y - 5)^\circ$, $\angle C = (4x)^\circ$ and $\angle D = (7x + 5)^\circ$. Find the four angles.
4. On selling a T.V. at 5% gain and a fridge at 10% gain, a shopkeeper gains ₹2000. But if he sells the T.V. at 10% gain and the fridge at 5% loss, he gains Rs.1500 on the transaction. Find the actual price of the T.V. and the fridge.
5. Two numbers are in the ratio 5 : 6. If 8 is subtracted from each of the numbers, the ratio becomes 4 : 5. Find the numbers.
6. 4 Indians and 4 Chinese can do a piece of work in 3 days. While 2 Indians and 5 Chinese can finish it in 4 days. How long would it take for 1 Indian to do it? How long would it take for 1 Chinese to do it?

3 Algebra_Term1.indd 133 02/12/2022 21:41:06